

**SARS-CoV-2 Relative**

A close relative of the COVID-19 causing pathogen has been found in frozen bats from Cambodia. <https://digit.in/dec20-23>

**Sobering up**

A study shows that a decent amount of alcohol detoxing can be achieved via hyperventilation. <https://digit.in/dec20-24>



HOW DO BICYCLES WORK?

Scientists still do not understand why a bicycle is so stable

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Tourists in China are told of a joke where a bicycle falls over because it is “two tired”. The funny thing about the joke is the inaccuracy, because the bicycle is surprisingly good at not falling over. When a person is riding a bicycle, they are taking cues from the environment and their own senses to make a number of tiny adjustments to keep the balance of the bicycle. This involves steering slightly in the direction you are going to fall, to

stabilise the bike. If the bike leans to the right, you steer right to prevent it from falling over. The funny thing is, that the bike seems to do this on its own, even when there is no one riding it. To check this out, all you have to do is push a bicycle on a flat surface and observe it. Eventually the bicycle does fall, but for a while it seems to steer itself in the direction that it is falling in, allowing it to remain stable. Researchers do not know how exactly the forces that stabilise the bike in its riderless condition play a role in the forces that stabilise the bike when it is being ridden by a person. Bicycle stability is thus an area of active scientific research, and it has many practical real world applications.

GYRO AND CASTER THEORIES

Another unintuitive thing about bicycles is that of countersteering,

where you momentarily have to steer to the left to initiate a turn to the right, or momentarily steer to the right to initiate a turn to the left, then follow through by steering and leaning to the left. Finally, there are all sorts of ways to control a bicycle without steering it at all. A person on a bicycle can maintain balance on a bike, even without touching the handlebars. It is also possible to manoeuvre a bike while riding superman style, with the stomach on the seat and both the arms and legs in the air. How can science explain these behaviors of a bicycle?

There are two basic theories to explain the stability of the bicycle. The first is the gyro theory, and the second is the caster theory. According to the gyro theory, the spinning motion of the wheel acts as a gyroscope, that causes the bicycle to stabilise itself. A dramatic demonstration of this effect can be seen by taking the front wheel off a bicycle, balancing the axle on a finger from either hand, and spinning the wheel. Once the wheel is in motion, even after a finger is removed, the wheel continues to turn on a single finger, till the circular motion can be maintained. If the wheel is not spinning, or stops spinning, it falls off the single finger. Make sure that the